The roots of the general sextic equation

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Abstract

In this paper, I describe the roots of the general sextic equation.
The paper ends with "The End"

Introduction

In a previous paper, I've described how to solve the general quintic equation. In a previous paper, I've described my monic sextic identity. In a previous paper, I've described how the roots of my monic sextic equation are expressible in radicals. In this paper, I describe the roots of the general sextic equation.

Preliminaries

The general sextic equation is

$$ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g = 0$$

where a, b, c, d, e, f, g are constants.

If $g = 0$ then the equation reduces to

$$x(ax^5 + bx^4 + cx^3 + dx^2 + ex + f) = 0$$

which has the root $x = 0$ and the quintic

$$ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$$

which can be solved.

Similarly, if $a = 0$ then the equation reduces to quintic equation

$$bx^5 + cx^4 + dx^3 + ex^2 + fx + g = 0$$

which can be solved.

If $a = g = 0$ then the equation reduces to

$$x(bx^4 + cx^3 + dx^2 + ex + f) = 0$$

which has the root $x = 0$ and the quartic

$$bx^4 + cx^3 + dx^2 + ex + f = 0$$

whose roots are known.

Therefore, $a \neq 0$ and $g \neq 0$ henceforth. Moreover, we divide the general sextic equation by the leading coefficient a to transform the general sextic equation to the monic sextic equation

$$x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$$

where $f \neq 0$ henceforth.

Comparing coefficients with my monic sextic

Recall that my monic sextic is

$$x^6 + ax^5 + bx^4 + \left((a-P)\left(b - aP + P^2 - Q\right) + PQ + \frac{eQ + fP - af}{Q^2} \right) x^3 + \left(\left(b - aP + P^2 - Q\right) Q + \frac{f}{Q} + \frac{(a-P)(eQ + fP - af)}{Q^2} \right) x^2 + ex + f$$

Comparing coefficients we get
(1) $a = a$
(2) $b = b$
(3) $c = (a-P)\left(b - aP + P^2 - Q\right) + PQ + \frac{eQ+fP-af}{Q^2}$
(4) $d = \left(b - aP + P^2 - Q\right) Q + \frac{f}{Q} + \frac{(a-P)(eQ+fP-af)}{Q^2}$
(5) $e = e$
(6) $f = f$

Thus, if suitable P and Q are obtained, then, by my monic sextic identity, we can reduce the monic sextic equation to the product of a quartic equation and a quadratic equation, whose roots are known.

Choosing P and Q

Eliminating f between equations (3) and (4) gives us the eliminant

(7) $a^4P - a^3b - 4a^3P^2 + a^3Q + 3a^2bP + a^2c + 6a^2P^3 - 6a^2PQ - 3abP^2 + 2abQ - 2acP - ad - 4aP^4 + 9aP^2Q - 2aQ^2 + bP^3 - 2bPQ + cP^2 - cQ + dP + e + P^5 - 4P^3Q + 3PQ^2 = 0$

As long as $Q \neq 0$ and we obtain corresponding P and Q, we may choose any value for either P or Q to solve the eliminant. For most sextics, $P = \frac{2a}{3}$ is a valid and convenient choice. When $P = \frac{2a}{3}$ is not a valid choice, other valid and convenient choices may be $P = 0$, $Q = P$ etc.

Solving the monic sextic

Once we have at least one valid value of P and one valid value of Q, by the right side of my monic sextic identity, we obtain 6 roots of the monic sextic equation.

Notes

1. Note that by following this procedure, we obtain 6 roots of the general sextic equation expressible in radicals.
2. Note that this procedure doesn't invalidate Galois' theory since our procedure is based on expressing the general sextic equation as **factored forms** but not on solving the sextic equation algebraically.

Exercises for the reader

Find the roots of the following sextic equations expressible in radicals:

1. $$x^6 + 506x^5 - 4932x^4 + 39002x^3 - 331779x^2 + 934280x - 440734 = 0$$

2. $$x^6 - 16x - 20 = 0$$

The End